

Problem

Design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of a parallel RLC circuit.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of a parallel RLC circuit.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A parallel RLC circuit has the following values:

$$R = 60 \, \Omega, \quad L = 1 \, \text{mH}, \quad \text{and} \quad C = 50 \, \mu\text{F}$$

Find the quality factor, the resonant frequency, and the bandwidth of the RLC circuit.

Solution

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 50 \times 10^{-6}}} = \underline{4.472 \, \text{krad/s}}$$

$$B = \frac{1}{RC} = \frac{1}{60 \times 50 \times 10^{-6}} = \underline{333.33 \, \text{rad/s}}$$

$$Q = \frac{\omega_o}{B} = \frac{4472}{333.33} = \underline{13.42}$$